

Situation

Gastroenterology fellows are currently assessed by a series of competency-based milestones established by the American Board of Internal Medicine throughout their three-year training. Unfortunately, these milestones remain rather vague with regards to definitions of endoscopic competency. There are a variety of endoscopic skill grading tools that have been developed by other institutions, but none are currently in routine use in our fellowship program. Fellows progress their endoscopic skills at varying rates and establishing a more effective means of assessment to provide feedback would be of great importance to better prepare them for independent practice.

Historically, competency used to be judged after completing a specific number of endoscopic procedures. Using the 2007 national core curriculum (Waxler, 2007), 130 upper endoscopies and 140 colonoscopies were recommended as a minimum prior to evaluating competence. Though these numbers have varied with subsequent curriculum updates or at non-US training programs, specific procedure volume numbers were subsequently removed in 2014 updates to the common core curriculum. As such, an improved rubric or similar assessment tool would be of great importance to better evaluate ipsative year-to-year progression and overall performance. Barriers towards implementation would include the added time to conduct such assessments along with training or buy-in from supervising attending physicians. No added costs are anticipated with such a change.

If this problem is not addressed, it can have far-reaching complications not only in fellowship but including post-graduation. In order for fellows to have hands-on experience, procedures often take longer than if the attending conducted the endoscopy on their own. Though the procedure times are often booked with some leeway, if fellows are behind in their endoscopic skills this can lead to procedural delays for patients or increased sedation times. As well, this can lead to a cascading effect where attendings have to take over procedures to limit delays, which therefore limits fellows from having hands-on endoscopy time (which is what they need to improve). If fellows are not competent to perform these procedures independently, this can affect their ability to graduate on time but can also lead to decreased confidence overall and negatively affect their independent practice in the future. If a fellow is behind compared to their peers and this is identified early, their schedule could potentially be adjusted to allow for more endoscopy opportunities or increased simulation time.

As such, the implementation of a more robust criterion-based assessment to better evaluate gastroenterology fellow endoscopic skills using a fixed-standard system would be of clear benefit. Further investigation is merited to evaluate established validated endoscopic technical assessment tools to determine how best to integrate into our current training scheme.

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Background

Colonoscopy is one of the key necessary skills for a gastroenterology fellow to master during their training in addition to upper endoscopies. Colonoscopies are often performed to screen for colon cancer and evaluate for polyps, with several criteria having been developed to determine competence (Rex et al., 2006). Some of these quality metrics include appropriate pre-procedure checklists (such as documentation of indication, appropriate holding of blood thinners), comment on preparation quality, minimum length of withdrawal time, cecum intubation rates, and detection rates for polyps. These skills are critical to impart onto fellows throughout their three-year gastroenterology fellowship.

Fellows starting gastroenterology fellowship typically have limited to no hands-on endoscopic background experience, as this is not covered in the prerequisite internal medicine residency training prior to fellowship. Whitson et al. (2022) describe four educational tools for endoscopic skill development: apprenticeship model, cognitive apprenticeship, coaching, and feedback, that had been the mainstay for training for decades. This however assumes that all endoscopic teachers are not only experts in endoscopy but also effective educators, which may not be the case. As such, apprenticeship can be much less effective as a result or cause variability in skill development between gastroenterology fellows depending on which attending they work with. To combat this, procedural numbers were often used as a benchmark to determine competency; however, this is not effective in clinical practice. Endoscopic procedures are not limited to gastroenterology training as these procedures are often required of colorectal surgery trainees and occasionally is included into select family medicine training programs. However, most endoscopies (and training) are still conducted by gastroenterologists nationwide. Beyond procedure number-based competency, there are no alternative standardized means of endoscopic procedure competency assessment incorporated into other specialties.

Alternate means of determining competency were therefore stressed by national gastroenterological societies. Until 2010, there was a dearth of effective assessment tools to provide competency-based feedback based on criterion-referenced standards established by Rex et al. (2006). Several tools have been developed since that time, though there remains inconsistent uptake across gastrointestinal programs including in our own gastroenterology fellowship at the University of Nebraska Medical Center. The implementation of an already validated tool at our institution would be of great benefit to not only provide facilitative feedback to fellows (Archer, 2009) to create a culture of effective feedback, but would also allow for early identification of learners that need corrective action or increased practical experience. If common weak themes were identified across fellows, this would lend itself well to improve competency-based education given its noted benefits of greater learner engagement, skill-based focus on educational objectives, and greater flexibility (Frank et al., 2010). An ideal tool would be cost-neutral to implement, not require significant additional training for evaluators, minimal time to complete, and should have already undergone testing to ensure reliability and validity (Banta and Palomba, 2014).

A literature review was therefore conducted to investigate already-developed endoscopic competency tools to assess gastroenterology fellows. A PubMed search with the search string “gastroenterology fellowship endoscopic scoring tool validated” identified fifteen publications. Simulation-based assessment tools were excluded as they would likely be costly to implement along with abstracts discussing advanced gastroenterology procedures (such as endoscopic ultrasound procedures) as these would not be applicable for a general gastroenterology fellowship. Ultimately, the Mayo

Colonoscopy Skills Assessment Tool (MCSAT), Assessment of Competency in Endoscopy tool (ACE), and Skill Assessment in Fellow Endoscopy Training (SAFE-T) scoring systems were selected for further review.

The MCSAT (Sedlack, 2010) was developed at the Mayo Clinic in Rochester, MN after evaluating gastroenterology fellows between 2007-2010. Validity for the tool was based on the work by Downing (2003) on developing education assessment validity frameworks. Domains in cognitive and motor competencies were incorporated, with a focus on colonoscopic skills. Upper endoscopic skills were not assessed using this tool. This is likely due to the fact that there are existing evidence-based metrics for colonoscopies for colon cancer screening purposes (which are not applicable to upper endoscopy), such as mucosa visualization, withdrawal times, and cecum intubation rates (Rex et al., 2006). Of note, sedation dosing was also included in the blueprint for this tool as many endoscopies at the time were performed using moderate sedation (i.e., sedation administered by the endoscopist) which is less relevant currently as much more procedures are conducted now with anesthesia support (monitored anesthesia care). To control for the variety of indications for conducting a colonoscopy, this scoring tool was evaluated on patients presenting for the same indication of colon cancer screening. For each of the thirteen criteria included in the tool, four scoring grades (and a non-applicable option) were included (novice, intermediate, advanced, superior). Each grading scale includes key descriptions to allow for accurate grading, and variability across assessments. There is no final percentage or letter grade attached to the completed assessment; however, both overall cognitive and hand-on skills are summarized using the same four scoring grades as described above (based subjectively on overall performance in those two over-arching domains).

Gastroenterology fellows were defined as novice, intermediate, or advanced in their skills using cutoffs of <20 procedures, 150 procedures, and 300 procedures respectively. Good correlation across all criteria ($p < 0.01$, correlation coefficients generally between 0.80-0.88 for cognitive domains) was seen when comparing MCSAT parameters against gastroenterology fellow procedural experience. Of note, during the study collection window, two fellows were identified to be performing below their peers (norm-referencing) and as a consequence their training schedule was adjusted to allow for two additional months of colonoscopy training to bring them on par. This tool has since been used to perform quarterly reviews of fellow colonoscopy performance compared against average learning curves, with training schedules adjusted where needed to provide more hands-on time. Sedlack goes on to describe this tool being used as a part of the review process when evaluating a graduating fellow interested in staying on as faculty after graduation (Sedlack, 2010). This tool has since been expanded to be used in a prospective multicenter trial (Sedlack, 2015) across ten academic institutions in the US with nearly identical findings to the benchmarks noted in the original 2010 paper and establishes minimal competency benchmarks based on this scoring tool.

As the MCSAT tool was developed for use for colonoscopies exclusively, it was later refined to the Assessment of Competency in Endoscopy tool (ACE) (Fried et al., 2013). This was split into two versions, the ACE-E for upper endoscopies, and the ACE-C for colonoscopies. The ACE-C is essentially the same as the MCSAT using the same cognitive and motor competencies for overall assessment, except with additional questions regarding polypectomy or cautery tool utilization, how independent was the fellow in tool use, and their knowledge of appropriate tool use and correct settings to be applied. The same scoring grades of novice, intermediate, advanced, and superior were maintained throughout the ACE-C, including in the additional tool utilization questions.

The ACE-E was also derived from the MCSAT as discussed above and also is based on the same two domains (cognitive and motor) for assessment. Compared to 13 questions for the ACE-C, the ACE-E comprises of 9 total questions and also includes the same questions for endoscopic intervention technique or tool use. The same metrics that are often applied to screening colonoscopies such as cecal intubations rate, polypectomy detection rate, and withdrawal time are not applicable to upper endoscopies and as such were not included for use. However, additional variables were included (intubation of the second portion of the duodenum, time to intubate second portion of the duodenum, and overall procedure time) to correlate against already established colonoscopy metrics (cecum intubation rate, withdrawal time) to aid in discriminating between trainees meeting competency minimums and not meeting metrics (as a mechanism to increase variability with the ACE-E tool). This scoring tool was later externally validated in a prospective multi-center trial across ten academic gastroenterology fellowship programs (Miller et al., 2019).

Finally, the Skill Assessment in Fellow Endoscopy Training tool (SAFE-T) was developed initially for colonoscopy use (Kumar et al., 2017) and later modified slightly to be applied to upper endoscopies (Kumar et al., 2020a). In both cases, this is a five-question assessment with the first question objectively documenting the anatomical landmark achieved, the second question regarding complexity of scope insertion (with three scoring grades of straightforward, average, challenging), and question three regarding the complexity of interventions (if performed). Finally, question four holistically summarizes the overall performance into five sub-grades (beginner, advanced beginner, intermediate, proficient, and superior) and question five allows the evaluator to select one area for the trainee to focus upon for their next procedure. The SAFE-T was validated using the same criteria as used during MCSAT development (Downing, 2003) and demonstrated excellent inter-rated reliability when tested using dual-observed colonoscopies (Kumar et al., 2017). The SAFE-T scoring tool was also developed with accessibility in mind, with a web-based tool created for point-of-care assessment. After initial development, the SAFE-T (colonoscopy) assessment was investigated in a multicenter prospective study which demonstrates not only consistent norm-referenced improvement year-on-year across three gastroenterology programs, but ease of implementation and generalizability given its five-question format (Kumar et al., 2020b).

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Assessment

To determine an optimal endoscopic skill assessment tool to be implemented into our gastroenterology fellowship program, one must evaluate the generalizability of the existing tools available, ease of use, time to complete while minimizing instructor training, and effectiveness in trainee evaluation. The tool should also not cost money to integrate and should be repeatable to track individual progress. Another description of an effective rubric is that it should be analytic, topic-specific, and completed with examples or rater training (Jonsson & Svingby, 2007). Based on the three assessment tools described above (MCSAT, ACE E/C, and SAFE-T) there are various potential benefits and drawbacks to each evaluation tool that impact their potential integration into our gastroenterology training program.

The MCSAT was the first tool developed specifically for endoscopic assessment for gastroenterology fellows; however, does have the notable drawback of only being applicable for colonoscopy use. This drawback is significant as an endoscopic assessment tool for gastroenterology fellows should include upper endoscopies as this is a core procedure that is required to demonstrate competence. Using the criteria created by Jonsson & Svingby, some objective criteria are implemented in the rubric though there is subjectivity through the Likert-type scale used for grading, and it is topic-specific as it was custom-created for endoscopic assessment (much like the other tools described below). Through its development, there are studies demonstrating internal and external validation (Sedlack, 2015). An effective rubric needs to be able to discriminate between learners of varying skills so that there is sufficient variance. The MCSAT is one such tool, with key differences in overall scoring accurately reflecting a trainee's degree of experience or the prior number of colonoscopies conducted (discriminating effectively between novice, intermediate, and advanced abilities) (Sedlack, 2010).

Building on the issues related to subjectivity in assessment, there are some vague areas that would merit further details (such as the extent reached without further anatomic details), along with the final 'global' assessment score for both cognitive and hands-on skills that does not incorporate a summative score, rather is subjective based on prior scoring elements. Also, as noted above, the inability of this scoring tool to be used for upper endoscopies does significantly limit its potential use. Based on my experience within our gastroenterology fellowship program, the colonoscopy-only use for this tool would be a barrier towards use in addition to the overall length of the survey which may pose a time limitation. Effective feedback is critical to medical education to foster "positive and desirable development" (Archer, 2009). The MCSAT does not allow for self-reflection, nor does it necessarily highlight specific areas for fellows to improve upon in future endoscopies – both of which would be important facets for a future integrated assessment tool in our gastroenterology fellowship.

As noted above, based on some of the illustrated drawbacks of the MCSAT, the ACE tool was created along a similar vein with more specific objective details included along with an adapted score for use in upper endoscopy skill assessment. Some specific improvements include an adaptation for use in upper endoscopies (ACE-E), and more specific questions regarding endoscopic tool use and therapeutic interventions. The same scoring grades as the MCSAT were utilized and otherwise covered largely similar questions overall. The ACE-E tool however does have potential drawbacks as noted by Kolb and Wagh (2019). Key upper endoscopic techniques (that do not have a specific corollary in colonoscopy technique) such as esophageal intubation or traversing the pylorus were not distinguished from each other and instead were lumped into one more generic criterion ("scope tip control/advancement techniques"). As

well, the use of duodenum intubation times and total procedure times are not accurate reflections of endoscopic competency nor are these times traditionally used in clinical practice. Based on provider preference, many opt to perform biopsies of the esophagus or the stomach before proceeding to duodenal inspection rather than after completing duodenal inspection. These specific time metrics may be more relevant in purely diagnostic endoscopies with no interventions, but then this would eliminate the value of including endoscopic tool use (such as cautery, or biopsy forceps) criteria. As such, adding clinically irrelevant details to an assessment tool would be questioned both by students and faculty and would not only detract from an effective feedback culture but would also decrease utilization as this would decrease confidence in the tool as a whole. The ACE tool, being derived from MCSAT, has similar length concerns and time-to-complete barriers. The ACE tool would provide increased objective feedback to learners with the added details regarding skillful endoscopic tool use that was not addressed in MCSAT. Similar to MCSAT, external validation was also conducted demonstrating not only validity but ability to be integrated into multiple gastroenterology fellowships nationwide (Miller et al, 2019). This would suggest that implementation would be feasible at Nebraska Medicine though the shortfalls as above are noted.

Finally, the SAFE-T tool provides a unique approach to assessment as compared to the MCSAT and ACE tools. Not only is the tool shorter, but it also provides more objective data regarding anatomical landmarks, the complexity of scope insertion and interventions, and a summative question categorizing performance into five categories (beginner, advanced beginner, intermediate, proficient, superior). There is also a fifth question describing a particular area for improvement or focus for trainees for future endoscopies. Though there are still some subjective elements, the addition of a feedback question is of particular utility for trainees. Feedback should be task-specific and conducted 'in action' (Archer, 2009), which is better embodied through this assessment tool. Though limiting the number of questions does beg the question of whether the tool is effective in discriminating effectively between students of varying skills, excellent inter-rater reliability was demonstrated during the testing/development of this tool (Kumar et al., 2017). As well the SAFE-T tool can also be utilized for either upper endoscopies or colonoscopies with only minor differences regarding anatomical landmarks. An additional benefit to the SAFE-T tool is that it was designed for point-of-care assessment through web-based tools. Though this can be important in portability, there is also an added potential complicating step through this. Faculty would not only need to know how to access the web-based tool (i.e., recall and use the URL) but there would also need to be a mechanism for data storage and student access to the results. This can be much more complicated than completing a simple paper-based tool; however, there is no reason why the SAFE-T has to remain web-based only and could be printed out for accessibility purposes. This would aid in integration within our fellowship, as creating a web-based tool for access and the need for a storage mechanism could otherwise add unnecessary cost and labor.

All three assessment tools are freely available to use and should take less than five minutes each to complete. However, the SAFE-T tool, being only five questions in length, does have significant time-based advantages compared to the MCSAT and ACE (which are, in essence, rather similar to each other). All three tools do not incorporate means for self-assessment or peer assessment – both of which could be advantageous to student learning (Jonsson & Svingby, 2007). Also, the lack of a summative score for all three assessments does limit the ability for a more detailed understanding of learner performance through a limited ability to incorporate classical test theory or investigate sources of variance. Despite this limitation, all three survey tools however still demonstrate the ability to effectively discriminate

between learners of varying skill mastery (a key requirement of an effective assessment tool) and each tool has been used in repeatable settings which provides important longitudinal feedback for learners (and instructors) and highlight if learners are performing below the expectation at their level of training.

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Recommendation

Based on the assessment tools described, i.e., MCSAT, ACE E/C, and SAFE-T, the SAFE-T assessment tool would be the most reasonable to integrate within our gastroenterology fellowship. Given the short time needed to complete, utility in both upper endoscopies and colonoscopies, and a targeted question directed towards specific endoscopic skill refinement, the SAFE-T also requires the least amount of time to train examiners and remains free-to-use. Unlike the original study that integrated this tool via an online mechanism (Kumar et al., 2017), given the logistical difficulties with tracking serial data, maintaining confidentiality, and variability of examiner comfort with technology – when integrating this tool, it would be best to start by having this tool completed manually on paper.

I propose using this assessment tool every six months throughout a fellow's 3-year training program. At each mark, it would be the fellow's responsibility to have five endoscopies total reviewed (ideally including both therapeutic upper endoscopies and colonoscopies using at least two different supervising attendings). By leveraging multiple examiners, this allows for more accurate assessments overall and better distinguishment between high and low performers through test-retest reliability (Drost, 2011). Through this proposed implementation, there is also the ability to track performance on both an individual and group basis. Given our training program recruits five fellows per year, the average scores per class could be used to follow group performance and serve as a norm-based reference (Renaissance, 2018) for individual comparison.

Prior to integration, it will be important to have input from current trainees, faculty, and program leadership regarding the proposed use of the SAFE-T tool and the six-monthly assessment frequency. This will allow time for modifications to be made to the proposed plan prior to implementation, allow for discussion to determine how the results will be used, and identify key data points to be tracked to allow for retrospective review of the implementation and track outcomes. This also allows for time for faculty to gain familiarity with the tool; with the SAFE-T being only five questions in length, I do not foresee any major barriers or delays from this perspective. With the need for baseline fellow and faculty input in mind, this should be achievable over the next six months to allow for integration starting with the 2025-2026 academic year (i.e., July 2025); therefore, the first set of assessments would take place in late 2025 (approximately December 2025). I have also already commenced early discussions with program leadership regarding the implementation of this tool, so the six-month implementation appears to be realistic. Keeping in mind the plan for six-monthly assessments, I foresee the need for at least one year of baseline data before soliciting further feedback from fellows and faculty regarding the use of the clinical tool and its means of implementation before adjusting as necessary. As noted above, though the original development and implementation of the SAFE-T tool was through a web-based assessment, I plan on having evaluators complete the assessment on paper (with records then to be scanned and digitally stored) for ease of implementation and reduce potential web development costs. The six-month interval assessments would also correspond to the already established six-monthly meeting of the clinical competency committee that already meets to review individual fellow progress through the training scheme, highlight skills that are being done particularly well, and provide constructive feedback in domains that need further refining (based on already established standardized graduate medical education criteria). The ACE-T tool would be an ideal tool for objective assessment of the 'Procedures: Cognitive Components' and 'Procedures: Technical Components' domains.

With regards to outcomes, if this intervention was successful, it would be of particular benefit in delivering objective feedback to trainees regarding endoscopic competencies and potentially be of importance for future employers of program graduates (to provide documented objective evidence of endoscopic proficiency rather than simply case volume). For trainees that are performing below expectations as determined by ACE-T, based on faculty discussion within the six-monthly meeting of the clinical competency committee, then a fellow's schedule over the next six months can be adjusted as necessary to include more opportunities for endoscopy. If a fellow is performing significantly below expectations, then the SAFE-T could also be repeated at a shorter interval (e.g., three months) for closer monitoring. Post-implementation, the results of the SAFE-T could also be used for internal validation and comparison against published national data. As well, the SAFE-T would be of particular benefit for a subset of fellows that integrate a second fellowship (transplant hepatology) within the same three years meant for gastroenterology fellowship. In this 2+1 arrangement, fellows are expected to complete the standard three-year requirements in only two years such that their third year is dedicated towards their transplant hepatology fellowship. By using the SAFE-T tool, this would add objective data in evaluating endoscopic competency to gauge if fellows have met the expected standard before moving onto their second fellowship.

In summary, the SAFE-T assessment tool fits the requirements of a robust criterion-based tool that shows clear benefits over the subjective assessments currently in use, and the objectivity will be of great benefit in better evaluating endoscopic skill competency for gastroenterology fellows. The ability to serially monitor individual/group progress across the 3-year training scheme will be invaluable during the already established six-monthly clinical competency group meetings held by training faculty, and the cost-neutral tool provides a low barrier towards implementation.

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